Proceedings of the 3rd International Conference on Sustainable Computing and Data Communication Systems (ICSCDS-2025)
DVD Part Number: CFP25AZ5-DVD; ISBN: 979-8-3315-1249-1

Adaptive Machine Learning Intrusion Detection System

Anand S
Department of Computer Science, RVS Technical Campus, Coimbatore, India.
anandsince2004@gmail.com

Deepak M
Department of Computer Science, RVS Technical Campus, Coimbatore, India.
priyanka.ka2024mecse@sece.ac.in

Vanitha M
Assistant Professor, Department of Computer Science, RVS Technical Campus, Coimbatore, India.
ananthi68@gmail.com

Vijetha T M
Department of Computer Science, RVS Technical Campus, Coimbatore, India.
vijethatm@gmail.com

Hariramakrishnan S
Department of Computer Science, RVS Technical Campus, Coimbatore, India.
oviyasivanandham@gmail.com

Abstract - The rapid evolution of cyber threats has made traditional security mechanisms insufficient for protecting modern networks. This research presents an Adaptive Machine Learning Intrusion Detection System (IDS) designed to detect and classify malicious network activities using intelligent data-driven techniques. The proposed system utilizes a two-stage machine learning pipeline that integrates anomaly detection and attack classification. Isolation Forest is used to identify suspicious network traffic, while Random Forest is used to classify attacks into categories such as Denial of Service (DoS), Probe, Remote to Local (R2L), and User to Root (U2R). The system is trained and evaluated using the NSL-KDD dataset consisting of more than 125,000 network traffic records.

Experimental results demonstrate that the proposed model achieves high detection accuracy with low false positive rates, making it suitable for real-time enterprise security monitoring systems.

Keywords: Intrusion Detection System, Machine Learning, Cybersecurity, Isolation Forest, Random Forest, Network Security

1. INTRODUCTION

Network security threats continue to grow rapidly with increasingly sophisticated attack techniques.

Traditional security solutions such as firewalls and signature-based detection systems are often unable to detect new or unknown attacks. Intrusion Detection Systems (IDS) provide an additional layer of protection by continuously monitoring network traffic and identifying suspicious activities.

However, conventional IDS approaches often struggle with high false positive rates and limited ability to detect zero-day attacks. Machine learning techniques provide a promising solution by

Machine learning–based intrusion detection systems are capable of analyzing large volumes of network data and identifying hidden patterns that may indicate malicious behavior. Unlike traditional rule-based systems, machine learning models can continuously improve their detection capability by learning from new data. Techniques such as classification, clustering, and anomaly detection help in distinguishing normal network behavior from potential attacks, thereby improving detection accuracy and reducing false alarms.

Furthermore, machine learning approaches enable IDS to detect previously unseen threats, including zero-day attacks, by identifying abnormal deviations in network activity. This makes them more flexible and scalable compared to conventional security solutions. With the advancement of artificial intelligence and data analytics, machine learning–driven IDS has become an active area of research in network security.

I. LITERATURE SURVEY

The rapid growth of network technologies has led to an increase in cyber threats, making intrusion detection an important area of research in network security. Numerous researchers have proposed different approaches to detect and prevent malicious activities in computer networks.

Denning [1] introduced one of the earliest models for intrusion detection systems based on anomaly detection. Her work laid the foundation for modern IDS by demonstrating how abnormal patterns in

breaches. However, the model relied heavily on predefined statistical thresholds, which limited its ability to adapt to evolving cyber threats.

Lippmann et al. [2] conducted extensive research on evaluating intrusion detection systems using the DARPA dataset. Their work provided a standardized framework for testing IDS performance and comparing different detection methods. Although the dataset helped researchers evaluate algorithms effectively, it was later criticized for not fully representing real-world network traffic.

Sommer and Paxson [3] examined the application of machine learning techniques in intrusion detection systems. Their study highlighted the advantages of machine learning in detecting unknown attacks but also pointed out challenges such as high false positive rates and difficulties in deploying models in real-world environments.

Tsai et al. [4] reviewed various machine learning approaches used in intrusion detection, including decision trees, neural networks, support vector machines, and clustering algorithms. Their comparative analysis showed that machine learning methods significantly improve detection accuracy compared to traditional signature-based systems. However, they noted that model performance strongly depends on the quality and balance of the training data.

Buczak and Guven [5] presented a comprehensive survey of data mining and machine learning techniques used in cybersecurity applications. Their research emphasized that combining multiple algorithms and feature selection techniques can improve the efficiency and accuracy of intrusion detection systems. Despite these improvements, scalability and real-time processing remain challenging issues.

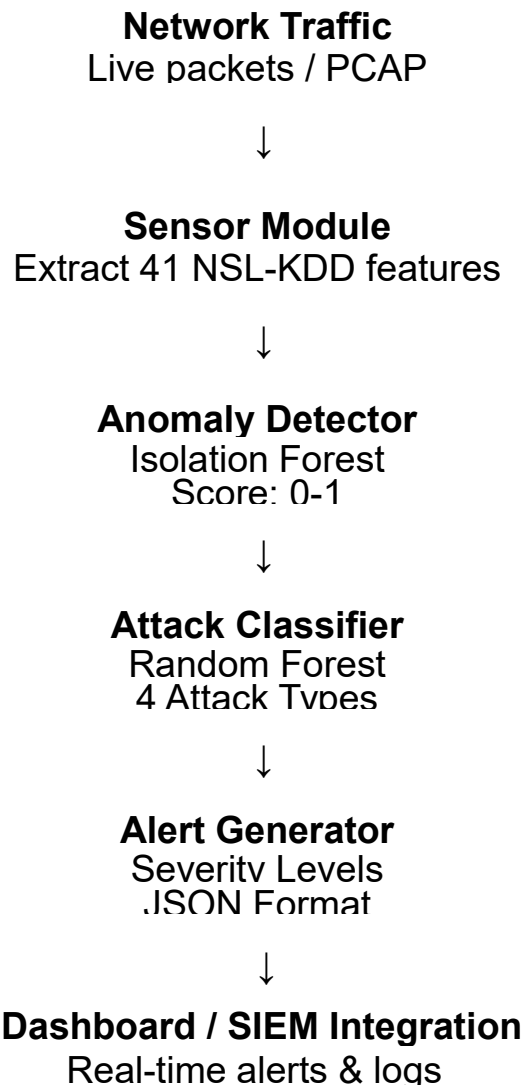
II. PROPOSED METHODOLOGY

The proposed system aims to develop an efficient **Intrusion Detection System (IDS)** using machine learning techniques to identify and classify malicious activities in network traffic. The methodology focuses on analyzing network data, extracting important features, and applying machine learning algorithms to detect potential cyber threats with improved accuracy.

Initially, a suitable **network traffic dataset** is collected from publicly available sources such as NSL-KDD or other intrusion detection datasets. This dataset contains various network parameters such as protocol type, duration, source and destination information, and other attributes that describe normal and malicious network behaviour. The collected data is then preprocessed to remove noise, handle missing values, and convert categorical attributes into numerical form suitable for machine learning models.

1) System Architecture Diagram

The proposed system architecture for the **Adaptive Machine Learning Intrusion Detection System (IDS)** is designed to monitor network traffic, detect suspicious activities, classify attacks, and generate real-time alerts.



The first component of the system is the **Threat Sensor Module**, which is responsible for capturing network traffic from the system or network environment. This module collects packets from network interfaces and extracts relevant features such as source IP address, destination IP address, protocol type, packet size, and connection flags.

The collected data is then forwarded to the **Intrusion Detection Module**, where machine learning algorithms analyze the network traffic to identify abnormal behavior. In this system, the Isolation Forest algorithm is used to detect anomalies in network traffic.

The **Attack Intelligence Module** is responsible for classifying the detected anomalies into specific types of cyber attacks. This module uses a Random Forest classifier that has been trained using network intrusion datasets. The classifier analyzes the suspicious traffic patterns and determines whether the activity corresponds to known attack categories such as Denial of Service (DoS), Probe attacks, Remote-to-Local (R2L) attacks, or User-to-Root (U2R) attacks

1) Thread Sensor Module

The first module in the system is the **Threat Sensor Module**, which acts as the data collection component of the intrusion detection system. This module continuously monitors the network environment and captures incoming and outgoing network packets from various network interfaces. The primary purpose of this module is to gather raw network traffic data and extract relevant features required for further analysis. Important attributes such as source IP address, destination IP address, protocol type, port numbers, packet size, and connection duration are collected from the captured packets. These extracted features represent the behavior of network traffic and help in identifying normal and abnormal activities. The collected data is then preprocessed and converted into a structured format that can be used by machine learning algorithms. This module ensures that the system receives accurate and real-time data for detecting potential threats in the network.


```
{
  "timestamp": "2026-01-21T00:35:00Z",
  "flow_id": "flow_001",
  "src_ip": "192.168.1.100",
  "dst_ip": "10.0.0.50",
  "src_bytes": 125000,
  "dst_bytes": 150,
  "protocol": "tcp",
  "syn_error_rate": 0.85,
  "features": {...41_features...}
}
```

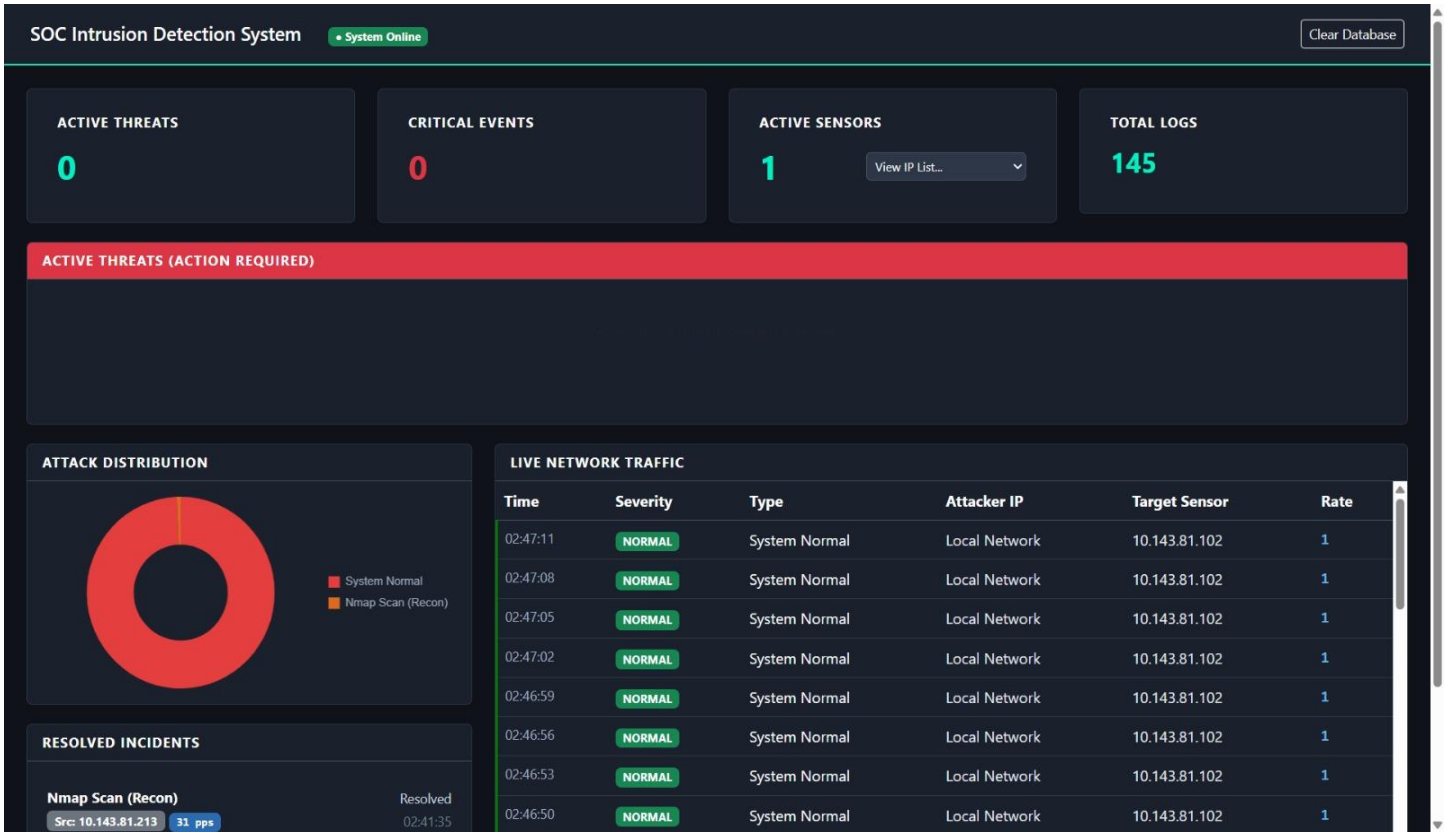

2) Intrusion Detection Module

The second module is the **Intrusion Detection Module**, which is responsible for analyzing the collected network data and identifying abnormal patterns. In this module, machine learning algorithms are used to detect anomalies in network traffic. The system compares incoming traffic patterns with previously learned normal behavior to identify deviations that may indicate potential attacks. In this project, the **Isolation Forest algorithm** is used for anomaly detection. Isolation Forest works by isolating data points that differ significantly from the majority of the data. If a network activity shows unusual behavior compared to normal traffic, it is marked as suspicious. This module calculates an anomaly score for each network instance and determines whether the activity is normal or potentially malicious. Suspicious traffic is then forwarded to the next module for further classification and analysis.


```
{
  "flow_id": "flow_001",
  "anomaly_score": 0.92,
  "is_suspicious": true,
  "deviation_metrics": {
    "src_bytes_deviation": "+450%",
    "connection_rate_deviation": "+320%"
  },
  "priority": "HIGH"
}
```

3) *Artificial Intelligence Module*

The third module is the **Attack Intelligence Module**, which is responsible for identifying and classifying the type of attack detected in the previous stage. Once suspicious network activities are identified, this module uses a trained machine learning classification model to determine the specific category of the attack. In this project, the **Random Forest algorithm** is used for attack classification. Random Forest is a supervised learning algorithm that uses multiple decision trees to improve classification accuracy. The module analyzes the features of suspicious traffic and compares them with known attack patterns present in the training dataset. Based on this analysis, it classifies the attack into categories such as **Denial of Service (DoS), Probe attacks, Remote-to-Local (R2L) attacks, and User-to-Root (U2R) attacks**. This classification helps system administrators understand the nature of the threat and take appropriate action to mitigate the attack.



III. FUTURE WORK

Although the proposed Adaptive Machine Learning Intrusion Detection System provides an effective method for detecting and classifying network intrusions, there are several areas where the system can be further improved in the future. One possible enhancement is the integration of advanced deep learning techniques such as neural networks and deep learning models to improve the accuracy and efficiency of intrusion detection. These models can better analyze complex patterns in network traffic and reduce false positive rates.

Another potential improvement is the implementation of real-time intrusion detection using streaming data from live networks. This would allow the system to detect attacks instantly and respond more quickly to potential threats. Additionally, the system can be expanded to support cloud-based environments and Internet of Things (IoT) networks, where security threats are increasing rapidly.

Future work may also include the development of an automated response mechanism that can automatically block malicious IP addresses or suspicious activities without requiring manual intervention. Furthermore, incorporating more diverse and updated datasets for training machine learning models can improve the system's ability to detect newly emerging cyber attacks. These enhancements will make the intrusion detection system more robust, scalable, and efficient in protecting modern network infrastructures.

IV. RESULT ANALYSIS AND PERFORMANCE

The result analysis of the proposed Adaptive Machine Learning Intrusion Detection System focuses on evaluating the effectiveness of the machine learning models used for detecting and classifying network intrusions. The system was tested using a network intrusion dataset containing both normal and malicious network activities. The dataset was divided into training and testing sets to evaluate the performance of the implemented algorithms.

The system first applies the Isolation Forest algorithm to detect anomalies in network traffic. This algorithm analyzes network behavior and identifies unusual patterns that deviate from normal traffic. Each network record is assigned an anomaly score, and records with higher anomaly scores are considered suspicious. The anomaly detection process helps filter potentially malicious activities from normal traffic efficiently.

After detecting suspicious activities, the system uses the Random Forest classifier to categorize the detected anomalies into specific attack types. The classifier analyzes the extracted network features and compares them with previously trained data to determine the type of attack. The system successfully classified different categories of attacks such as Denial of Service (DoS), Probe attacks, Remote-to-Local (R2L), and User-to-Root (U2R). This classification process helps security administrators understand the nature of the attack and take appropriate preventive measures.

The performance of the system was evaluated based on key metrics such as accuracy, detection rate, and false positive rate. The Random Forest model achieved high classification accuracy while maintaining a relatively low false positive rate. The Isolation Forest algorithm effectively detected anomalous traffic, enabling the system to identify suspicious activities at an early stage. The combination of anomaly detection and classification techniques improves the overall efficiency of the intrusion detection process.

Overall, the results demonstrate that the proposed Adaptive Machine Learning Intrusion Detection System provides reliable and efficient detection of network intrusions. The use of machine learning algorithms enhances the ability of the system to identify complex attack patterns and improves the overall security of the network environment.

V. CONCLUSION

The **Adaptive Machine Learning Intrusion Detection System** is developed to enhance network security by identifying and classifying malicious activities in network traffic. With the rapid growth of internet usage and connected devices, networks have become more vulnerable to various cyber threats. Traditional security systems are often unable to detect new and evolving attacks effectively. Therefore, the proposed system uses machine learning techniques to improve the detection of network intrusions.

The system collects network traffic data, analyzes it using anomaly detection algorithms, and classifies suspicious activities into different attack categories.

© May 2026 | JIESURT | Volume 12 Issue 12 | ISSN: 2278-0183

The use of algorithms such as Isolation Forest for anomaly detection and Random Forest for attack classification helps improve the accuracy and efficiency of the detection process. By identifying abnormal patterns in network behavior, the system can detect potential threats at an early stage.

Overall, the proposed system provides a reliable and efficient solution for detecting cyber attacks in network environments. It helps administrators monitor network activity, identify security threats, and take necessary actions to prevent damage. The implementation of machine learning techniques makes the system adaptable to new attack patterns, thereby improving overall network security.

REFERENCES

[1] M. Tavallaee, E. Bagheri, W. Lu, and A. A. Ghorbani, "A Detailed Analysis of the KDD CUP 99 Data Set," *Proceedings of the IEEE Symposium on Computational Intelligence for Security and Defense Applications*, 2009.

[2] L. Breiman, "Random Forests," *Machine Learning Journal*, vol. 45, no. 1, pp. 5–32, 2001.

[3] F. T. Liu, K. M. Ting, and Z. H. Zhou, "Isolation Forest," *IEEE International Conference on Data Mining (ICDM)*, 2008.

[4] S. Axelsson, "Intrusion Detection Systems: A Survey and Taxonomy," *Technical Report*, Chalmers University of Technology, Sweden, 2000.

[5] W. Stallings and L. Brown, *Computer Security: Principles and Practice*, 4th ed., Pearson Education, 2018.

[6] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*, MIT Press, 2016.

[7] Scikit-learn Developers, "Scikit-learn: Machine Learning in Python," Available: https://scikit-learn.org

[8] NSL-KDD Dataset, "Network Intrusion Detection Dataset," Available: https://www.unb.ca/cic/datasets/nsl.html

© May 2026 | JIESURT | Volume 12 Issue 12 | ISSN: 2278-0183